# Ekjot Brar

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## Education

**University of Alberta** Edmonton, AB
*Bachelor of Applied Science, Minor in Applied Mathematics* *Sep. 2023 – May 2028*

## Experience

**Artificial Intelligence Research Assistant** May 2026 – August 2026
*University of Alberta* *Edmonton, AB*

- Researched neuro-fuzzy and machine learning techniques for intelligent decision-making systems, performing model development, testing, and performance analysis.
- Developed Python-based data processing and evaluation pipelines, supporting reproducible experimentation and quantitative assessment of AI algorithms.

**Service Router Platform Test Dev** Jan. 2026 – May 2026
*Nokia* *Ottawa, ON*

- Engineered and scaled modular SR/SR-L router topologies, improving re-usability and efficiency of network validation frameworks.
- Developed Python-based automation pipelines in a Linux (Ubuntu) regression environment for carrier-grade router hardware/software validation.
- Built dynamic Precision Time Protocol (PTP) test environments with parameterized configurations (IP, VLAN, interface speeds) to accelerate automated test setup.
- Conducted system-level validation of synchronization technologies (PTP, GNSS, SyncE), ensuring accurate timing and clock alignment across network devices.
- Designed object-oriented Python abstractions mapped to YANG data models for real-time configuration and monitoring of central frequency clock systems.

**Information Technology Support Specialist** May 2025 – August 2025
*Tetra Tech* *Edmonton, AB*

- Supported field deployment and troubleshooting of advanced pavement inspection systems (LiDAR, LCMS, laser sensors) ensuring reliable high-resolution data collection.
- Developed a QR-based tracking system for hard drives using structured database workflows to improve data traceability and inventory management.
- Processed and validated pavement datasets using MATLAB, identifying anomalies (bridges, bumps) and improving data quality assurance workflows.
- Developed C/C++-based tools for configuration and data handling of imaging systems (Ladybug camera), supporting integration into analysis pipelines.

## Projects

**ECO Car** | *Python, C/C++, KiCAD, STM32IDE* May 2025 – Present

- Contributed to Shell Eco-Marathon hydrogen vehicle development, gaining hands-on experience with embedded systems and low-power electronics relevant to semiconductor applications.
- Designed and routed a 2-layer PCB in KiCad, including schematic capture, component selection, and layout optimization for efficient electronic system integration.
- Programmed an STM32 microcontroller for real-time control and PWM-based LED operation, building embedded C and digital control experience.
- Assembled and soldered SMD/through-hole components, supporting hardware prototyping and circuit validation.
- Implemented FPGA digital logic using VHDL, including finite state machines and combinational/sequential circuit design.

**AlbertaSAT** | *ANSYS Mechanical, SolidWorks, AutoCAD, Git, MATLAB* May 2018 – May 2020

- Contributed to the development and microgravity testing of a gecko-adhesive debris capture system (CAN-RGX), focusing on system-level performance validation under dynamic and structural constraints
- Performed vibrational and structural analysis of CubeSat subsystems, evaluating modal response, stress distribution, and mechanical stability of critical components
- Supported Ex-Alta 3 CubeSat hardware development, including hinge mechanism design, CAD modeling, and investigation of UHF integration with emphasis on mechanical-electrical system reliability

## Technical Skills

**Languages**: C/C++, Python, MATLAB, Java, SQL, HTML/CSS, R
**Embedded & Hardware**: STM32, FPGA, VHDL, KiCAD, Altium, Oscilloscope, Logic Analyzer
**Developer Tools**: Linux (Ubuntu), Git, VS Code, Visual Studio, MATLAB
**Design & Simulation**: KiCad, SolidWorks, ANSYS Mechanical, AutoCAD